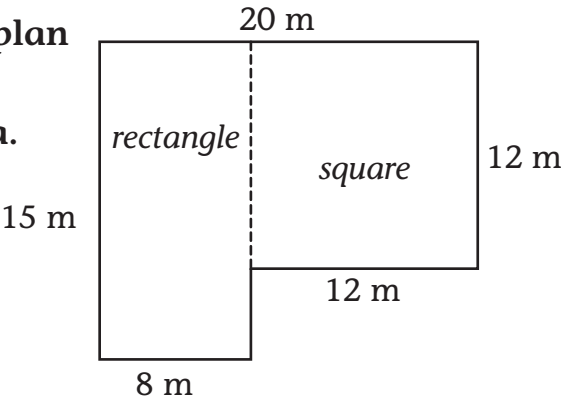


2.4 Areas of composite shapes

A composite shape is made up from two or more simpler shapes.

EXAMPLE

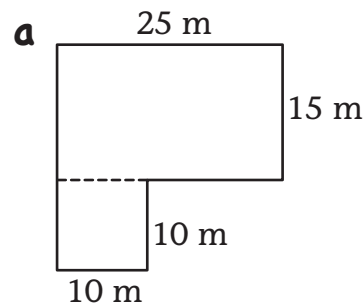
Here is the floor plan of a house.
Calculate its area.



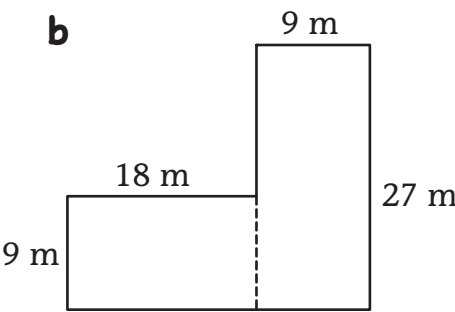
The dotted line shows how the figure is made from two shapes.

$$\begin{aligned} \text{Area} &= \text{rectangle} + \text{square} \\ &= 15 \times 8 + 12 \times 12 \\ &= 120 + 144 \\ &= 264 \text{ m}^2 \end{aligned}$$

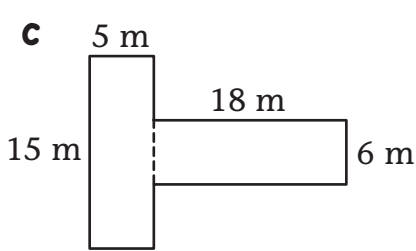
- 1 Calculate the area of each composite figure.
Dotted dividing lines are shown. They divide the shapes into rectangles.



$$\begin{aligned} \text{Area} &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \end{aligned}$$

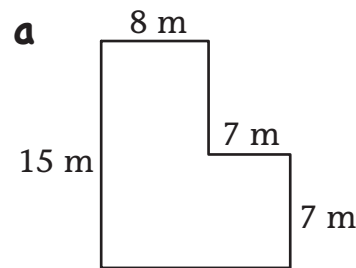


$$\begin{aligned} \text{Area} &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \end{aligned}$$

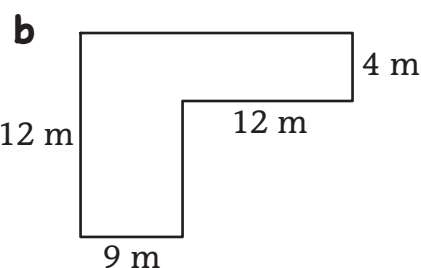


$$\begin{aligned} \text{Area} &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \end{aligned}$$

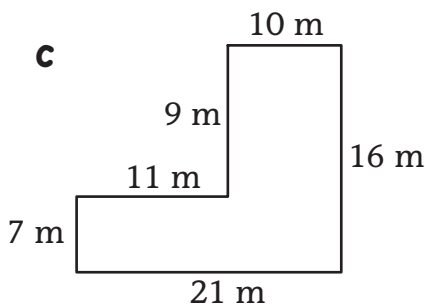
- 2 Draw lines to divide each shape into rectangles.
Then find the total area of each shape.



$$\begin{aligned} \text{Area} &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \end{aligned}$$



$$\begin{aligned} \text{Area} &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \end{aligned}$$

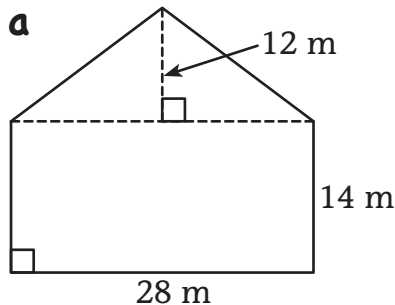


$$\begin{aligned} \text{Area} &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \end{aligned}$$

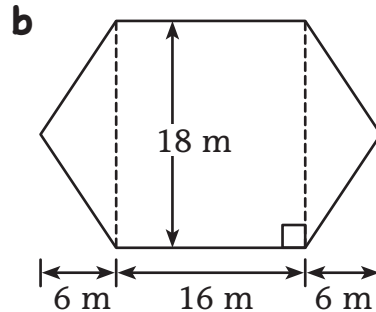
2.4 Areas of composite shapes

CONTINUED

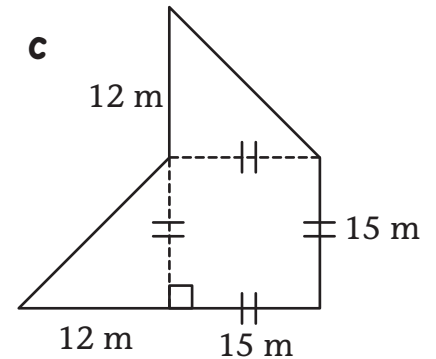
- 3** The following shapes are made up of rectangles, squares and triangles. Calculate the total area of each shape.



Area = _____
 = _____
 = _____



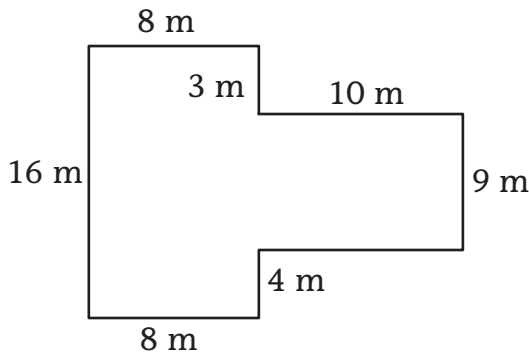
Area = _____
 = _____
 = _____



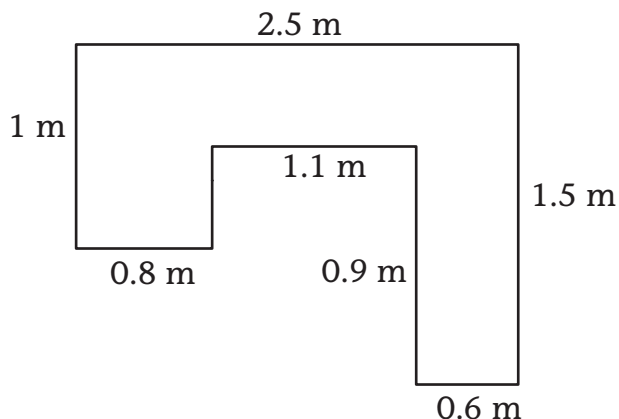
Area = _____
 = _____
 = _____

Answer the following questions in your workbook.

- 4** The diagram shows the plan of a house. Calculate its area.



- 5** The top of a kitchen bench has this shape. Calculate its area.



- 6** A farm consists of three rectangular paddocks. What is the total area of the farm:

- a** in square metres?
b in hectares?
 (1 hectare = 10 000 m²)

